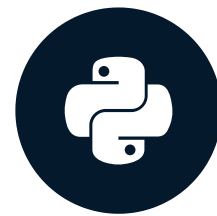


Load up and look at some data

PYTHON FOR R USERS



Daniel Chen
Instructor

Overview

- 2013 NYC Flights
- Load and Explore
- Manipulate and plot

Importing data with list comprehensions

- List comprehension to load data
- Saves repetitive typing
- Condensed way of appending values to a list

List comprehensions

```
import glob
import pandas as pd
csv_files = glob.glob('*.csv')
csv_files
```

```
['data3.csv', 'data2.csv', 'data1.csv']
```

```
all_dfs = [pd.read_csv(x) for x in csv_files]
all_dfs[0]
```

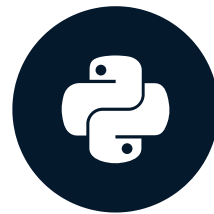
	A	B	C	D
0	a0	b0	c0	d0
1	a1	b1	c1	d1
2	a2	b2	c2	d2
3	a3	b3	c3	d3

Let's practice!

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Manipulating data

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Groupby

```
df = pd.DataFrame({
    'name': ['John Smith', 'Jane Doe', 'Mary Johnson'],
    'treatment_a': [np.NaN, 16, 3],
    'treatment_b': [2, 11, 1]
})
df_melt = pd.melt(df, id_vars='name')
df_melt.groupby('name')['value'].mean()
```

```
name
Jane Doe      13.5
John Smith     2.0
Mary Johnson   2.0
Name: value, dtype: float64
```


Groupby aggregate

```
df_melt.groupby('name')['value'].agg(['mean', 'max'])
```

	mean	max
name		
Jane Doe	13.5	16.0
John Smith	2.0	2.0
Mary Johnson	2.0	3.0

Dummy variables

- Categorical variables need to be encoded as dummy variables
- One-hot encoding

```
df = pd.DataFrame({  
    'status': ['sick', 'healthy', 'sick'],  
    'treatment_a': [np.NaN, 16, 3],  
    'treatment_b': [2, 11, 1]  
})  
df
```

	status	treatment_a	treatment_b
0	sick	NaN	2
1	healthy	16.0	11
2	sick	3.0	1

Get dummies

```
pd.get_dummies(df)
```

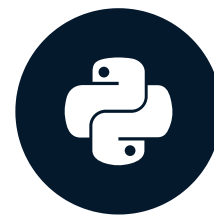
	treatment_a	treatment_b	status_healthy	status_sick
0	NaN	2	0	1
1	16.0	11	1	0
2	3.0	1	0	1

Let's practice!

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Wrap-up

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Review of topics

- How R translates into Python
- Basic types, functions, methods
- Numpy and Pandas
- Data Manipulation and cleaning techniques
- Visualization

R vs Python

- One language isn't "better" than the other
- Broaden your toolkit

Let's practice!

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